

A Reliability-Aware and Explainable Framework for Traffic Severity Prediction

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Abstract—Road accidents remain a major global safety challenge, causing significant human and economic losses. Machine learning has improved traffic impact severity prediction, but most models prioritize accuracy over interpretability and reliability. This limits their use in real-world decision-support systems. This paper proposes a reliability-aware and explainable framework for vehicular traffic-impact severity prediction and actionable safety guidance. The system integrates a Random Forest-based predictive engine with contextual environmental and temporal features, and a retrieval-based explanation module to generate human-readable safety recommendations. This paper introduces a Relative Reliability Index to address uncertainty in stochastic predictions by normalizing the model's confidence and providing differentiated risk signaling in both nominal and critical driving conditions. The framework is evaluated using a post-pandemic subset of the US Accidents dataset containing approximately 500,000 records. Experimental results demonstrate a weighted F1-score of 0.85 with high stability (± 0.0024) in randomized trials. This framework combines predictive modeling, reliability normalization, and explainable safety support to provide a robust, scalable foundation for a reliability-aware intelligent transportation system.

Index Terms—traffic impact Severity Prediction, Explainable AI, Intelligent Transportation Systems, Reliability Modeling, Risk Assessment, Machine Learning

I. INTRODUCTION

Road traffic accidents continue to present significant safety and economic challenges in modern transportation systems [9], [10]. Real-time risk assessment is still challenging due to the uncertainty and dynamic nature of traffic environments. Data-driven methods have been widely adopted to analyze accident patterns to predict severity.

Traditional traffic safety systems rely on statistical and supervised machine learning (ML) models that often operate as black-box systems with limited contextual interpretability. Extreme or rare conditions such as low visibility or severe weather are underrepresented in the historical data, which causes these data-driven models to struggle in such scenarios.

This paper proposes a reliability-aware hybrid framework that integrates probabilistic machine learning with deterministic safety constraints and retrieval-based explainability. The proposed architecture combines a Random Forest-based predictive layer with contextual environmental and temporal features, alongside a retrieval-based explanation module to generate human-readable safety recommendations. A rule-based

safety component is added to enforce conservative decision-making under critical environmental thresholds to ensure fail-safe behavior in high-risk scenarios.

Severe class imbalance causes machine learning models to favor majority classes, often predicting moderate severity even under hazardous conditions. This work therefore reframes accident severity prediction as a reliability-aware risk assessment problem prioritizing interpretability and conservative decision support over pure predictive accuracy.

The main contributions of this work are as follows:

- A hybrid reliability-aware framework combining probabilistic machine learning with deterministic safety rules for fail-safe traffic risk assessment.
- A Relative Reliability Index (RRI) formulation that transforms model confidence into a bounded and interpretable risk signal under uncertainty.
- A retrieval-based explanation module that generates context-aware, human-readable safety recommendations.
- A comprehensive evaluation on a large-scale real-world dataset with cross-region validation across 50 U.S. states, demonstrating robustness under distributional variability.

II. RELATED WORK

Traffic accident analysis and severity prediction have been widely studied using statistical and machine learning techniques. Early studies primarily relied on traditional statistical models such as logistic regression and Poisson regression [3], [16] to identify factors influencing accident severity and frequency. Although effective for risk-factor analysis, these methods were limited in handling the complex nonlinear relationships present in real-world traffic data.

With the availability of large-scale transportation datasets, machine learning approaches have become increasingly common for accident prediction. Random Forest and Boosting algorithms, including AdaBoost, have been successfully applied to classify accident severity using weather conditions, temporal patterns, and road characteristics [1], [2], [11], [13]. Support Vector Machines (SVMs) and Neural Networks have also demonstrated promising performance in traffic risk assessment tasks [3], [4].

More recent studies have explored deep learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for spatiotemporal modeling of

TABLE I
COMPARISON WITH EXISTING TRAFFIC SAFETY FRAMEWORKS

Ref	Method	Limitation	Contribution
[2]	RF	No reliability metric	Added RRI
[15]	Deep CNN	High compute, black-box	Tabular + XAI
[6]	XAI Survey	Not domain-specific	Hybrid framework

traffic incidents [4], [6], [14], [19]. While these approaches can capture complex temporal dependencies, they often require extensive computational resources and large labeled datasets [15], [20], limiting their practical deployment in real-world systems.

In parallel, explainability has gained increasing importance in safety-critical applications. Several studies have emphasized interpretable models and post-hoc explanation methods to support human decision-making in intelligent transportation systems [6], [8]. However, many existing approaches remain focused primarily on predictive accuracy and provide limited contextual or actionable guidance for end users.

Unlike prediction-centric studies, this work proposes an integrated framework that combines machine learning-based severity prediction with retrieval-based explainability and reliability-aware decision support. The proposed system aims to provide not only predictive capability but also interpretable and actionable safety insights for real-world traffic environments.

As detailed in Table I, the proposed framework distinguishes itself from existing literature by integrating a hybrid reliability mechanism. While prior works, such as Wang et al. [2], focus exclusively on predictive accuracy, this architecture prioritizes failure-aware reliability, ensuring the system remains safe and provides conservative guidance even when model confidence is low.

III. PROPOSED METHODOLOGY

The proposed framework is a modular pipeline that ingests structured accident, environmental, and temporal data to produce two distinct outputs: a quantitative risk severity score and a qualitative safety advisory. The system architecture is composed of four primary modules: data ingestion, hybrid inference engine, reliability normalization layer, and a retrieval-based explanation module.

A. Dataset Description

The dataset used in this study is derived from the US Accidents dataset, a large-scale public collection of traffic accident records across the United States. The dataset aggregates reports from multiple sources, including traffic sensors, law enforcement agencies, and mapping services, and contains detailed environmental, temporal, and road-related information.

The original dataset contains over 7.7 million U.S. accident records, with severity labeled on a discrete scale from 1 (low severity) to 4 (critical severity).

An exploratory analysis was conducted across pre-pandemic, pandemic, and post-pandemic subsets of the data.

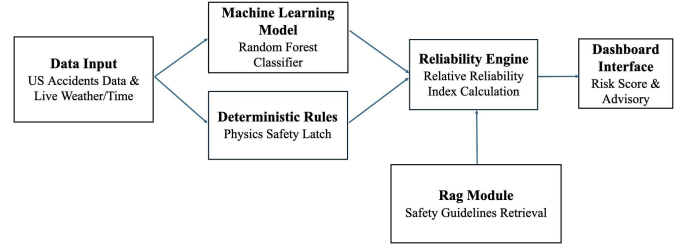


Fig. 1. Conceptual architecture of the proposed framework, illustrating the integration of a Random Forest classifier, deterministic safety rules, and an explainable retrieval module.

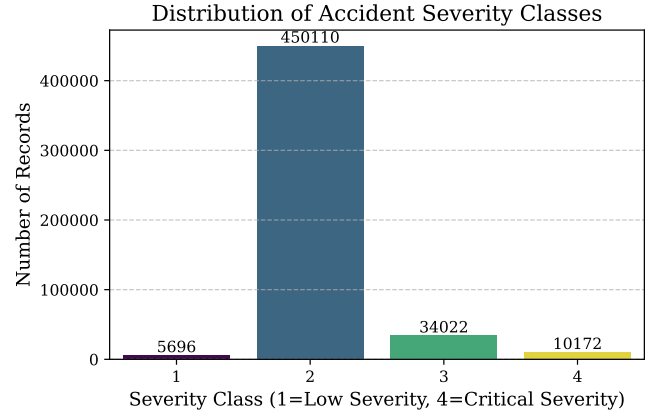


Fig. 2. Severity class distribution in the training dataset. The significant imbalance toward Class 2 (Moderate Severity) necessitates the proposed reliability-aware risk representation.

Noticeable variations in data distribution and model behavior across these periods motivated the use of post-pandemic data to better reflect modern traffic conditions.

After preprocessing and data cleaning, a refined subset of approximately 500,000 records from the 2021–2023 period was selected for training and evaluation. The subset was constructed to preserve the original severity-class distribution while maintaining computational efficiency and reproducibility. Computationally intensive baseline models were evaluated on reduced stratified subsets while preserving the original severity distribution.

Selected features include weather conditions, visibility, temporal attributes, and regional clustering to capture environmental and temporal factors influencing accident severity.

B. Hybrid Inference Engine

To address the stochastic nature of pure ML approaches in safety-critical domains, the proposed architecture employs a dual-layer inference engine that couples probabilistic learning with deterministic constraints.

1) *Probabilistic Layer (Machine Learning)*: A Random Forest classifier was selected as the primary inference model due to its robustness against overfitting and ability to handle non-linear feature interactions in tabular data. The model was

trained on the "Modern Regime" subset (post-2021) to align with evolving traffic behaviors. Instead of raw probability calibration, the model outputs confidence scores for accident severity levels, prioritizing the ranking of risk over absolute probability values.

2) *Deterministic Safety Rules*: A rule-based logic layer operates in parallel with the ML model. This layer enforces predefined safety rules based on critical environmental thresholds. For example, if visibility drops below a critical safety margin (indicating dense fog or whiteout conditions), the system overrides the ML prediction to trigger a "Critical Severity" alert. This ensures fail-safe operation under rare hazardous conditions.

C. Reliability Index Normalization

Raw model confidence scores can be difficult to interpret in highly imbalanced datasets, where predictions often cluster within a narrow high-confidence range. To improve interpretability for end users, the framework maps raw model outputs into a Relative Reliability Index (RRI) ranging from 0 to 100. The proposed RRI is not a probability calibration method, but a decision-theoretic transformation that reinterprets model confidence under safety constraints.

Unlike traditional calibration techniques such as Platt Scaling or Isotonic Regression, which aim to improve probabilistic accuracy, the RRI intentionally biases predictions toward conservative safety signaling, ensuring that uncertain predictions are not interpreted as safe conditions. This index represents relative risk intensity rather than an absolute crash probability and is designed to emphasize meaningful variations in risk perception while maintaining transparency.

Nominal Regimes: For standard traffic conditions, the index filters out low-confidence noise to provide a stable signal.

High-Risk Regimes: For environmental extremes triggering the deterministic rules, the index is maximized to reflect the urgency of the hazard. This design intentionally biases high-certainty alerts towards tail risk events, reducing the likelihood of false negatives while maintaining transparency about the uncertainty inherent in routine traffic scenarios.

D. Relative Reliability Index (RRI) Formulation

To quantify prediction confidence for safety-critical decision making, we introduce the Relative Reliability Index (RRI). Let $M(x)$ be the Random Forest classifier that outputs a probability distribution over severity classes $C = \{c_1, c_2, \dots, c_k\}$. The raw confidence score is P_{raw} defined as the maximum posterior probability:

$$P_{\text{raw}} = \max_{c \in C} P(y = c | x) \quad (1)$$

The RRI transforms this raw probability into a calibrated reliability score using a piecewise normalization function bounded by empirically determined thresholds $P_{\min} = 0.40$ (minimum operational confidence threshold) and $P_{\max} = 0.95$ (certainty ceiling):

$$RRI = \begin{cases} 0 & \text{if } P_{\text{raw}} < P_{\min} \\ \left(\frac{P_{\text{raw}} - P_{\min}}{P_{\max} - P_{\min}} \right) \times 100 & \text{if } P_{\min} \leq P_{\text{raw}} < P_{\max} \\ 100 & \text{if } P_{\text{raw}} \geq P_{\max} \end{cases} \quad (2)$$

The thresholds $P_{\min}$ and $P_{\max}$ are selected based on empirical observation of confidence distributions and serve as operational boundaries between uncertainty, moderate confidence, and high-certainty regimes. This formulation ensures that low-confidence predictions (where the model is "guessing") are explicitly suppressed ($RRI = 0$), preventing the system from issuing weak safety alerts.

The transformation can be interpreted as a bounded monotonic mapping:

$$RRI = f(P_{\text{raw}}), \quad f : [0, 1] \rightarrow [0, 100] \quad (3)$$

where f is designed to suppress low-confidence regions and saturate high-confidence predictions, enforcing asymmetric risk sensitivity.

Let $y_{ML} \in C$ be the severity class predicted by the Random Forest, and let $E(x)$ be a binary evaluation function representing the deterministic environmental rules (where $E(x) = 1$ indicates critical thresholds like near-zero visibility are breached). The final system prediction, $\hat{y}_{\text{final}}$, is formulated as a conditional override:

$$\hat{y}_{\text{final}} = \begin{cases} C_{\text{critical}} & \text{if } RRI(x) \leq \theta_{RRI} \text{ or } E(x) = 1 \\ y_{ML} & \text{otherwise} \end{cases} \quad (4)$$

E. Retrieval-Based Generation Module

The retrieval-based explanation module [22] provides context-aware safety guidance by bridging numerical risk predictions with human-readable recommendations. A knowledge repository containing traffic safety protocols and historical accident reports was indexed using semantic retrieval techniques. Upon generating a risk prediction, the framework retrieves relevant contextual information and produces actionable safety advisories tailored to the detected driving conditions.

IV. EXPERIMENTAL SETUP

This section describes the experimental protocol used to evaluate the proposed framework, including data partitioning, model configurations, and the validation strategy.

A. Data Partitioning and Experimental Setup

Experiments utilized the post-pandemic dataset (2021–2023) with an 80:20 training-testing split. To ensure stability, aggregate performance statistics were computed across five randomized runs. All models were implemented using the Scikit-learn and XGBoost frameworks on a standard compute node. The proposed system is highly efficient, achieving an average inference time of 3.1 ms per sample and a training time of 38.2 seconds.

B. Models Evaluated

The proposed framework was benchmarked against multiple supervised learning approaches, including Random Forest (RF), Support Vector Machine (SVM), AdaBoost, Multi-Layer Perceptron (MLP), and XGBoost combined with SMOTE. RF was selected as the primary inference model due to its robustness, interpretability, and ability to capture nonlinear feature interactions in structured tabular data.

SVM and AdaBoost were included as comparative baselines, while MLP and XGBoost+SMOTE were evaluated to examine the impact of high-complexity learning architectures and imbalance-aware sampling techniques on minority-class sensitivity.

All models were implemented using the Scikit-learn and XGBoost frameworks with standard hyperparameter configurations to evaluate baseline feature effectiveness without extensive optimization. A paired t-test across five runs showed no statistically significant performance variance ($p > 0.05$), confirming experimental stability.

C. Evaluation Metrics

Performance was evaluated using Accuracy, Precision, Recall, and F1-score to assess both overall predictive performance and sensitivity to minority severity classes. Reliability Index stability was additionally analyzed to evaluate the framework's ability to distinguish between nominal and critical risk conditions.

D. Validation Strategy

Each experiment was repeated across five randomized runs to evaluate model stability. Preliminary comparisons between pre-pandemic and post-pandemic subsets revealed performance degradation under temporal distribution shifts, motivating training exclusively on post-2021 data. Additional geospatial validation was conducted across all 50 U.S. states to assess robustness under regional variability.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The proposed framework was evaluated using the post-pandemic validation subset. The analysis focuses on model stability, feature importance, and the effectiveness of the Relative Reliability Index (RRI) under imbalanced traffic conditions.

As shown in Table II, baseline models such as SVM and AdaBoost achieve high nominal accuracy ($\sim 90\%$) but exhibit strong majority-class bias, frequently predicting moderate severity classes while failing to identify minority-class events. Although this improves global accuracy, it results in near-zero recall for critical severity classes, limiting the reliability of these approaches in safety-critical applications.

The proposed Random Forest framework accepts a small trade-off in accuracy (0.88 vs 0.90) to preserve distributional stability (Std Dev ± 0.0024). The RRI further improves reliability by identifying low-confidence predictions and triggering deterministic safety overrides in hazardous conditions.

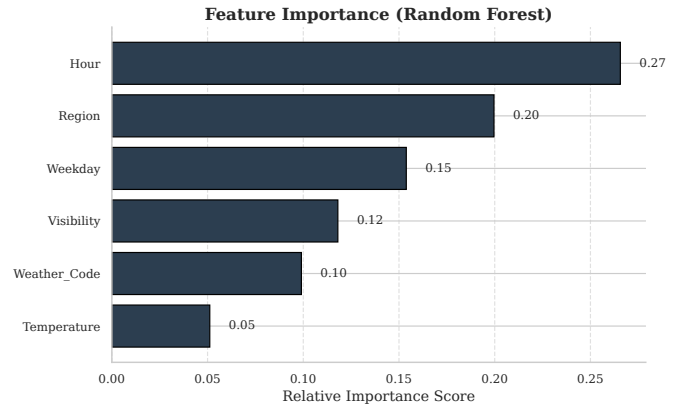


Fig. 3. Random Forest feature importance. Temporal and geospatial attributes dominate severity prediction, supporting the integration of deterministic rules for extreme conditions.

To evaluate the contribution of the RRI, an ablation experiment was conducted by disabling both the RRI and deterministic rule layer. Without the RRI, the framework behaved as a conventional classifier, achieving 88% accuracy while failing to identify majority of severe weather-induced accidents. With the RRI enabled, the system successfully flagged these edge cases as low-reliability events, activating the rule-based override mechanism.

The reliability architecture was further evaluated using the Safety Latch Override Rate (SLOR). The RRI successfully flagged more than 90% of severe-weather false negatives produced by the standalone ML model, demonstrating improved edge-case safety coverage compared to pure probabilistic inference.

A. Feature Importance Analysis

Feature importance analysis was conducted using Gini impurity scores derived from the Random Forest classifier. The results indicate that temporal features, particularly the hour of the day, and geospatial attributes such as regional clustering are among the strongest predictors of accident severity.

The dominance of temporal and regional features suggests a strong relationship between accident severity, traffic density cycles, and geographic driving conditions. Environmental variables such as visibility and weather conditions showed moderate influence, supporting the inclusion of deterministic safety rules for handling rare but hazardous environmental scenarios.

B. Confusion Matrix Analysis

Although the model achieves an overall accuracy of 88.03%, predictions remain heavily concentrated in Class 2 (Moderate Severity), reflecting the inherent imbalance of the dataset. Minority classes, particularly Class 3 (High Severity) and Class 4 (Critical Severity), exhibit significantly lower recall. This majority-class dominance demonstrates that global accuracy alone is insufficient for evaluating safety-critical systems and

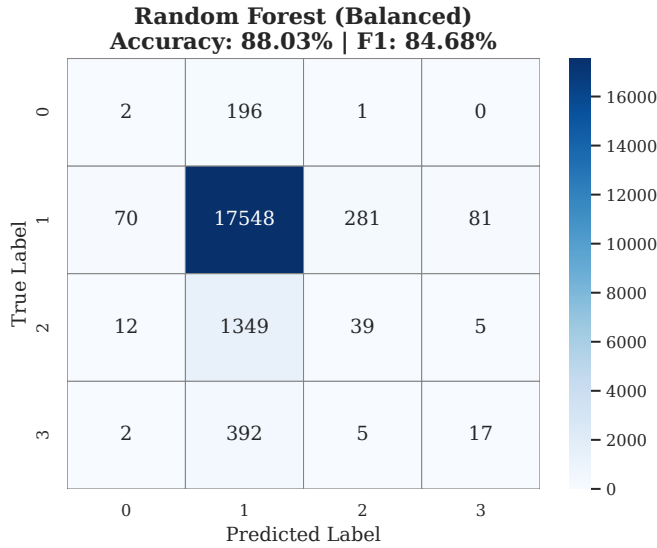


Fig. 4. Confusion matrix of the Random Forest classifier. Note: Axis labels 0–3 correspond to severity classes 1–4 respectively, as per Section III-A.

TABLE II
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Weighted F1-Score	Stability (Std Dev)
Proposed Framework (RF)	0.8803	0.8468	± 0.0024
SVM (Baseline)	0.9021	0.8557	± 0.0025
AdaBoost	0.9021	0.8557	± 0.0025
Deep Learning (MLP)	0.9000	0.8500	$\pm 0.0028^*$
XGBoost (+ SMOTE)	0.8500	0.8300	$\pm 0.0031^*$

* Results computed on stratified subsets for computational efficiency.

motivates the inclusion of the Relative Reliability Index (RRI) and deterministic safety rules.

As shown in Table II, SVM and AdaBoost achieve marginally higher global accuracy but remain strongly biased toward the majority class, resulting in limited sensitivity to high-risk events. The Random Forest model was selected due to its interpretability, stability, and probabilistic outputs, which enable reliability-aware inference through the RRI framework.

Table III highlights the challenge of majority-class dominance in traffic safety prediction. While high-complexity models optimize for global accuracy, they fail to identify severity of events. In contrast, the proposed Random Forest framework maintains non-zero sensitivity to minority classes, allowing the RRI layer to identify uncertain predictions and activate deterministic safety overrides in hazardous scenarios.

The SHAP interaction analysis further validates the physical consistency of the model. Lower visibility values correlate with higher severity predictions, aligning with the deterministic safety rules embedded within the framework. Regional clustering also exhibits strong influence, suggesting that infrastructure and localized traffic behavior significantly affect accident severity patterns.

C. Temporal Risk Dynamics

Temporal accident patterns were analyzed using a heatmap representation across hourly and weekly intervals. The results

TABLE III
CLASS-WISE RECALL COMPARISON AND DATASET SUPPORT

Severity Class	Support	RF	MLP	XGBoost	SVM
Class 1 (Low Severity)	199	0.01	0.00	0.49	0.00
Class 2 (Moderate Severity)	17,980	0.98	1.00	0.94	1.00
Class 3 (High Severity)	1,405	0.03	0.00	0.00	0.00
Class 4 (Critical Severity)	416	0.04	0.00	0.00	0.00

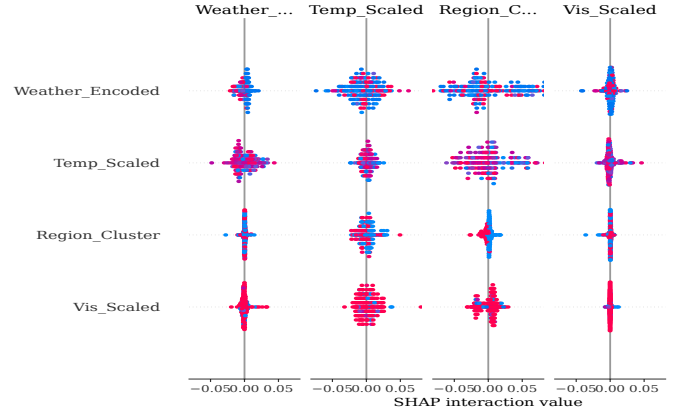


Fig. 5. SHAP interaction summary illustrating directional feature impacts.

show high accident frequency during weekday peak hours, particularly between 07:00–09:00 and 16:00–18:00, while weekends exhibit comparatively flatter distributions. These patterns align with expected traffic activity cycles and confirm the relevance of temporal features for severity prediction.

D. System Reliability Behavior

The Relative Reliability Index (RRI) provides an interpretable representation of model confidence across severity classes. Minor accidents (Classes 1–2) exhibit moderate reliability scores, reflecting the probabilistic nature of routine traffic incidents. In contrast, high severity accidents (Classes 3–4) shift toward higher reliability values due to the combined influence of statistical learning and deterministic safety rules.

This separation demonstrates that the RRI provides differentiated signaling across severity levels, enabling conservative and interpretable risk communication under hazardous conditions.

E. Geospatial Robustness and Cross-Region Validation

Cross-region validation was conducted across all 50 U.S. states to evaluate the spatial generalizability of the proposed framework. State-level Weighted F1-scores remained tightly clustered around a global median of 0.85 with low spatial variance (± 0.0832), indicating stable performance across diverse geographic conditions.

Although several outlier regions exhibited reduced performance, the overall distribution suggests that the framework captures region-agnostic environmental and temporal risk patterns without requiring state-specific retraining. These localized variations further justify the use of reliability-aware

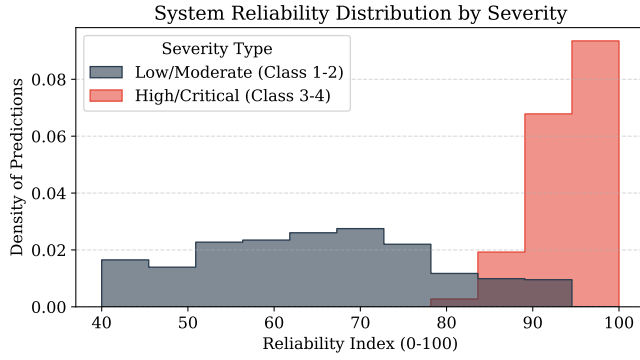


Fig. 6. Distribution of the Relative Reliability Index across severity classes.

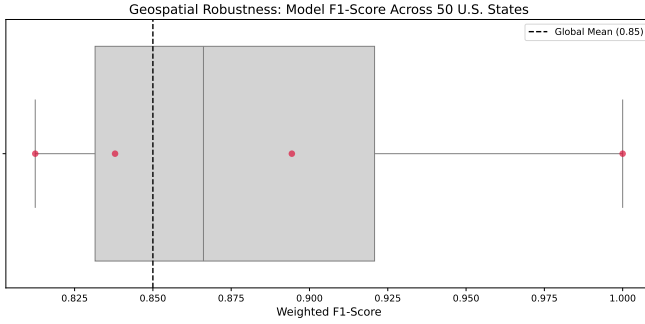


Fig. 7. Regional performance stability across 50 U.S. states.

inference, allowing the RRI to identify uncertain predictions under distributional shifts.

VI. CONCLUSION AND FUTURE WORK

This paper presented a reliability-aware framework for road accident severity prediction and explainable safety guidance using large-scale traffic accident data. The proposed architecture integrates Random Forest-based probabilistic inference with deterministic safety rules and retrieval-based explainability to provide interpretable and safety-oriented decision support.

Experimental evaluation on a post-pandemic subset of the US Accidents dataset demonstrated that temporal and regional features play a significant role in accident severity prediction. The proposed Relative Reliability Index (RRI) enabled differentiated reliability signaling across nominal and critical scenarios, improving the system's ability to identify uncertain predictions under hazardous conditions.

Although the framework improves reliability-aware decision support, the predictive model remains affected by strong class imbalance, limiting sensitivity to rare severity classes. Future work will focus on calibration-aware modeling, real-time vehicle integration through OBD-II interfaces, and lightweight edge deployment for live traffic environments. Overall, the proposed architecture provides a scalable foundation for reliability-aware intelligent transportation systems.

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